their vocabulary and articles. However, as we shall see in the following chapters, the whole edifice of modern politics and ethics is built upon subjective experiences, and few ethical dilemmas can be solved by referring strictly to brain activities. For example, what's wrong with torture or rape? From a purely neurological perspective, when a human is tortured or raped certain biochemical reactions happen in the brain, and various electrical signals move from one bunch of neurons to another. What could possibly be wrong with that? Most modern people have ethical qualms about torture and rape because of the subjective experiences involved. If any scientist wants to argue that subjective experiences are irrelevant, their challenge is to explain why torture or rape are wrong without reference to any subjective experience.

Finally, some scientists concede that consciousness is real and may actually have great moral and political value, but that it fulfils no biological function whatsoever. Consciousness is the biologically useless by-product of certain brain processes. Jet engines roar loudly, but the noise doesn't propel the aeroplane forward. Humans don't need carbon dioxide, but each and every breath fills the air with more of the stuff. Similarly, consciousness may be a kind of mental pollution produced by the firing of complex neural networks. It doesn't do anything. It is just there. If this is true, it implies that all the pain and pleasure experienced by billions of creatures for millions of years is just mental pollution. This is certainly a thought worth thinking, even if it isn't true. But it is quite amazing to realise that as of 2016, this is the best theory of consciousness that contemporary science has to offer us.

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Maybe the life sciences view the problem from the wrong angle. They believe that life is all about data processing, and that organisms are machines for making calculations and taking decisions. However, this analogy between organisms and algorithms might mislead us. In the nineteenth century, scientists described brains and minds as if they were steam engines. Why steam engines? Because that was the leading technology of the